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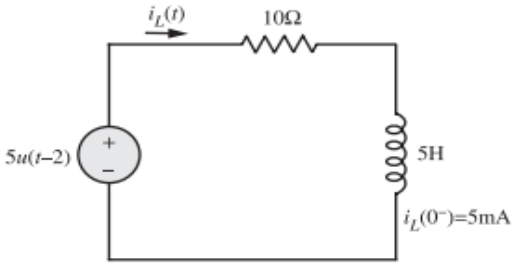
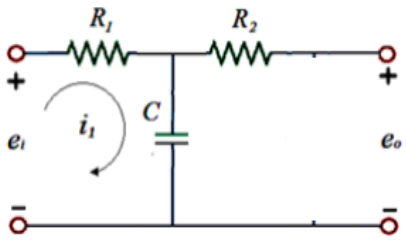
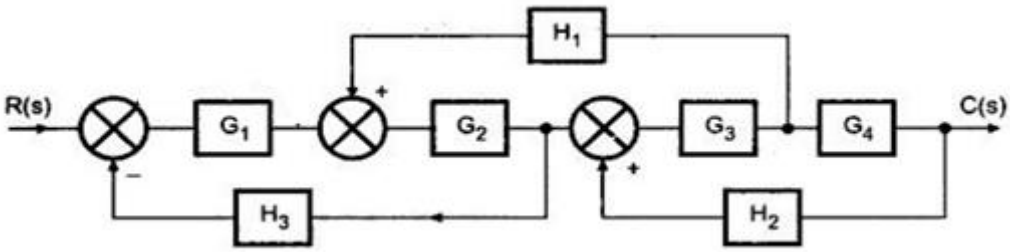
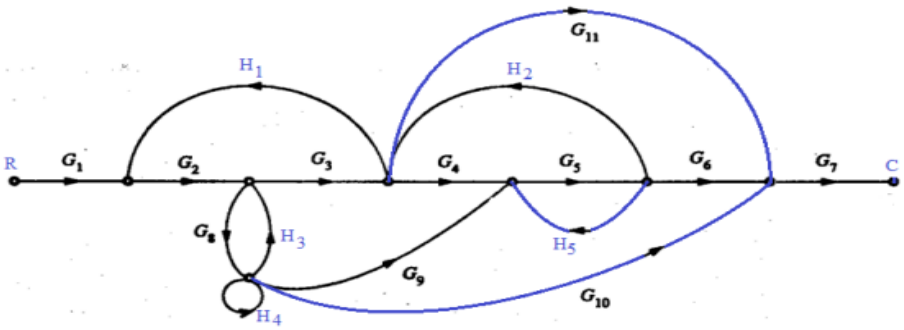
RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

NBA Accredited (UG - 6 Years)

Department of Electronics & Communication Engineering

Date: 24.02.2024	Test - 2	Max. Marks:50
Semester: III	UG	Duration: $1\frac{1}{2}$ Hrs
Course: Network Analysis and Control Engineering		Code:EC235AT

S No	Questions	M	BT	CO
1. A	Referring to the circuit of Figure. 1a write a differential equation for the inductor current $i_L(t)$. (b) Find $I_L(S)$ the Laplace transform of $i_L(t)$. (c) Solve for $i_L(t)$.  Figure Q. 1a  Figure Q.4b	06	3	3
1.b	Differentiate between Open Loop and Closed Loop Control system with examples.	04	1	3
2.	Determine the Transfer Function $\frac{C(S)}{R(S)}$ using Block Diagram Reduction Technique for the block diagram shown in Figure Q.2  Figure Q.2	10	3	3
3.	Determine the Transfer Function $\frac{Y5}{y1}$ for the signal flow graph shown in Figure Q3.a) using Mason's Gain Formula.  Figure Q3.a)	10	4	3
4. A	A second order system is given by $\frac{C(S)}{R(S)} = \frac{25}{S^2+6S+25}$. Find its Rise time, peak time, peak overshoot if subjected to unit step input. Also calculate expression for its output response.	06	3	4
4. b	Find the Transfer Function $\frac{Vo(S)}{Vi(S)}$ for the Electrical system shown in Figure Q.4b)	04	3	4
5.	Determine the step response of Second order system. Also plot the response of the system	10	4	4

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Semester: III	**UG**	**Duration: $1\frac{1}{2}$ Hrs**
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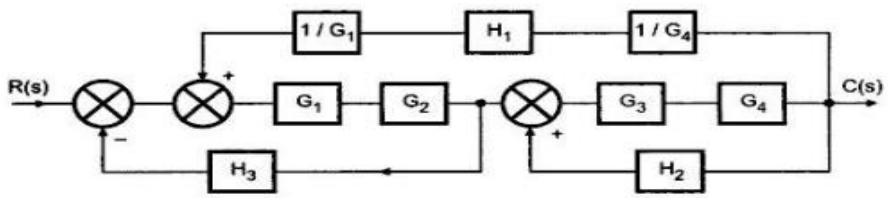
SCHEME & SOLUTIONS

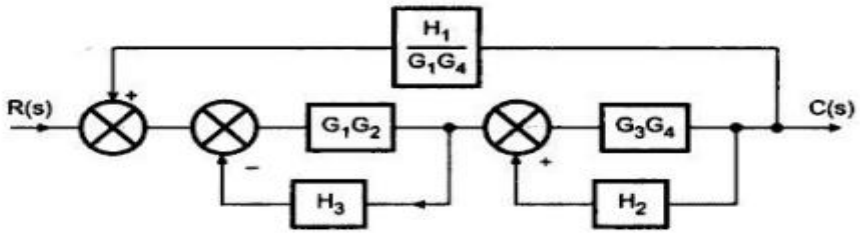
S No	Solutions with Scheme	Marks
1. A	(a) *Applying KVL clockwise*, we get $$10i_L(t) + 5\frac{di_L}{dt} - 5u(t-2) = 0$$ (b) Taking Laplace transform of the above equation, we get $$10I_L(s) + 5[sI_L(s) - i_L(0^-)] = \frac{5}{s}e^{-2s}$$ $$\Rightarrow \quad I_L(s) = \frac{\frac{5}{s}e^{-2s} + 5i_L(0^-)}{5s+10} = \frac{e^{-2s} + 5\times 10^{-3}s}{s(s+2)} = e^{-2s}\left[\frac{K_1}{s} + \frac{K_2}{s+2}\right] + \frac{5\times 10^{-3}s}{s(s+2)}$$ where $K_1 = \left.\frac{1}{s+2}\right\|_{s=0} = \frac{1}{2}$ $K_2 = \left.\frac{1}{s}\right\|_{s=-2} = -\frac{1}{2}$ Hence, $$\boldsymbol{I_L(s) = \frac{1}{2}e^{-2s}\left[\frac{1}{s} - \frac{1}{s+2}\right] + \frac{5\times 10^{-3}}{(s+2)}}$$ (c) Taking Inverse Laplace transform, we get $$i_L(t) = \frac{1}{2}\left[u(t) - e^{-2t}u(t)\right]_{t\to t-2} + 5\times 10^{-3}e^{-2t}u(t)$$ $$\boldsymbol{= \frac{1}{2}\left[u(t-2) - e^{-2t}u(t-2)\right] + 5\times 10^{-3}e^{-2t}u(t)}$$ a) 2M b) 2M c) 2M	06
B		04

	Open Loop		Closed Loop
1)	Any change in output has no effect on the input i.e. feedback does not exists.	1)	Changes in output, affects the input which is possible by use of feedback.
2)	Output measurement is not required for operation of system.	2)	Output measurement is necessary.
3)	Feedback element is absent.	3)	Feedback element is present.
4)	Error detector is absent.	4)	Error detector is necessary.
5)	It is inaccurate and unreliable.	5)	Highly accurate and reliable.
6)	Highly sensitive to the disturbances.	6)	Less sensitive to the disturbances.
7)	Highly sensitive to the environmental changes.	7)	Less sensitive to the environmental changes.
8)	Bandwidth is small.	8)	Bandwidth is large.
9)	Simple to construct and cheap.	9)	Complicated to design and hence costly.
10)	Generally are stable in nature.	10)	Stability is the major consideration while designing
11)	Highly affected by nonlinearities.	11)	Reduced effect of nonlinearities.

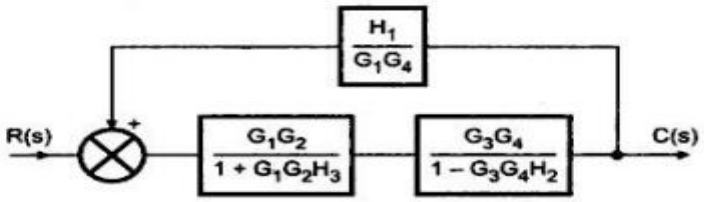
2. **Sol. :** Shifting take off point after the block 'G_4' and summing point behind the block having transfer function G_1 simultaneously, we get, — 10

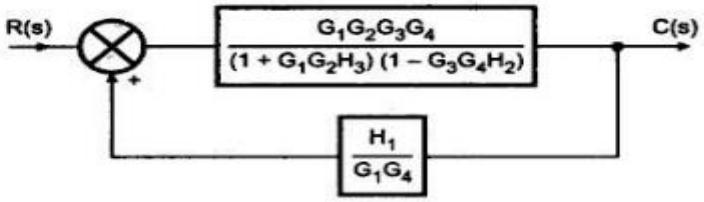


2M

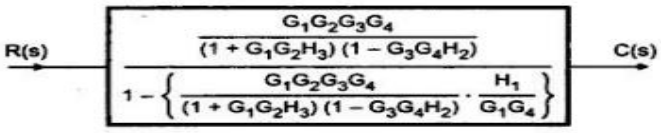


Solving both minor feedback loops we get,





5M



After simplification

$$\frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3 G_4}{(1+G_1 G_2 H_3)(1-G_3 G_4 H_2)-G_2 G_3 H_1}$$

3M

3.

There are FIVE forward path gains which are: 10M

P1 = G1 G2 G3 G4 G5 G6 G7 P2 = G1 G2 G3 G11 G7 P3 = G1 G2 G8 G9 G5 G6 G7

P4 = G1 G2 G8 G9 G5 H2 G11 G7 P5 = G1 G2 G8 G10 G7 1M

There are SIX individual loops,

the gains of these loops are

L1 = G8H3

L2 = H4

L3 = G2 G3H1

L4 = G4G5H2

L5 = G2G8G9G5H2H1

L6 = G5 H5 3M

There are SIX pairs of non-touching loops, the gains of these loops are

L1L4 = G8H3 G4G5H2

L1L6 = G8H3 G5 H5

L2L3 = H4 G2 G3H1

L2L4 = H4 G3G5H2

L2L6 = H4 G5 H5

L3L6 = G2 G3H1 G5 H5 3M

There is only ONE three non-touching loops, the gains of this loops are
L2L3L6 = H4 G2 G3H1 G5 H5 1M

$$\Delta_1 = 1 - L_2$$

$$\Delta_2 = 1 - \{L_2 + L_6\} + \{L_2 L_6\}$$

$$\Delta_3 = \Delta_4 = 1$$

$$\Delta_5 = 1 - L_6 - L_4$$

$$\Delta = 1 - \{L_1 + \cdots + L_6\} + \{L_1 L_4 + L_1 L_6 + \cdots + L_3 6\} - \{L_2 L_3 L_6\}$$

$$\frac{C(S)}{R(S)} = \frac{P_1\Delta_1 + P_2\Delta_2 + P_3\Delta_3 + P_4\Delta_4 + P_5\Delta_5}{\Delta}$$

2M

4. A Rise time:0.554sec, 1M 06
Peak time: 0.79sec 1M
% overshoot:9.48% 2M
C(t)=1-1.125 e(-3t) sin(4t+53.13°) 2M

b. $\frac{Vo(S)}{Vi(S)}$ =R/(R(LC^{2}S^3+1)) …..2M KVL to loop:2M 04

5. 10

We can modify the denominator term of the transfer function as follows –

$$s^2 + 2\delta\omega_n s + \omega_n^2 = \left\{s^2 + 2(s)(\delta\omega_n) + (\delta\omega_n)^2\right\} + \omega_n^2 - (\delta\omega_n)^2$$

$$= (s + \delta\omega_n)^2 + \omega_n^2(1 - \delta^2)$$

The transfer function becomes,

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{(s + \delta\omega_n)^2 + \omega_n^2(1 - \delta^2)}$$

$$\Rightarrow C(s) = \left(\frac{\omega_n^2}{(s + \delta\omega_n)^2 + \omega_n^2(1 - \delta^2)}\right) R(s)$$

Substitute, $R(s) = \frac{1}{s}$ in the above equation.

$$C(s) = \left(\frac{\omega_n^2}{(s + \delta\omega_n)^2 + \omega_n^2(1 - \delta^2)}\right)\left(\frac{1}{s}\right)$$

$$= \frac{\omega_n^2}{s\left((s+\delta\omega_n)^2+\omega_n^2(1-\delta^2)\right)}$$

Do partial fractions of $C(s)$.

$$C(s) = \frac{\omega_n^2}{s\left((s+\delta\omega_n)^2+\omega_n^2(1-\delta^2)\right)} = \frac{A}{s} + \frac{Bs+C}{(s+\delta\omega_n)^2+\omega_n^2(1-\delta^2)}$$

After simplifying, you will get the values of A, B and C as $1,\ -1\ and\ -2\delta\omega_n$ respectively. Substitute these values in the above partial fraction expansion of C(s).

$$C(s) = \frac{1}{s} - \frac{s+2\delta\omega_n}{(s+\delta\omega_n)^2+\omega_n^2(1-\delta^2)}$$

$$C(s) = \frac{1}{s} - \frac{s+\delta\omega_n}{(s+\delta\omega_n)^2+\omega_n^2(1-\delta^2)} - \frac{\delta\omega_n}{(s+\delta\omega_n)^2+\omega_n^2(1-\delta^2)}$$

$$C(s) = \frac{1}{s} - \frac{(s+\delta\omega_n)}{(s+\delta\omega_n)^2+(\omega_n\sqrt{1-\delta^2})^2} - \frac{\delta}{\sqrt{1-\delta^2}}\left(\frac{\omega_n\sqrt{1-\delta^2}}{(s+\delta\omega_n)^2+(\omega_n\sqrt{1-\delta^2})^2}\right)$$

Substitute, $\omega_n\sqrt{1-\delta^2}$ as ω_d in the above equation.

$$C(s) = \frac{1}{s} - \frac{(s+\delta\omega_n)}{(s+\delta\omega_n)^2+\omega_d^2} - \frac{\delta}{\sqrt{1-\delta^2}}\left(\frac{\omega_d}{(s+\delta\omega_n)^2+\omega_d^2}\right)$$

Apply inverse Laplace transform on both the sides.

$$c(t) = \left(1 - e^{-\delta\omega_n t}\cos(\omega_d t) - \frac{\delta}{\sqrt{1-\delta^2}}e^{-\delta\omega_n t}\sin(\omega_d t)\right)u(t)$$

$$c(t) = \left(1 - \frac{e^{-\delta\omega_n t}}{\sqrt{1-\delta^2}}\left((\sqrt{1-\delta^2})\cos(\omega_d t) + \delta\sin(\omega_d t)\right)\right)u(t)$$

If $\sqrt{1-\delta^2} = \sin(\theta)$, then 'δ' will be cos(θ). Substitute these values in the above equation.

$$c(t) = \left(1 - \frac{e^{-\delta\omega_n t}}{\sqrt{1-\delta^2}}(\sin(\theta)\cos(\omega_d t) + \cos(\theta)\sin(\omega_d t))\right)u(t)$$

$$\Rightarrow c(t) = \left(1 - \left(\frac{e^{-\delta\omega_n t}}{\sqrt{1-\delta^2}}\right)\sin(\omega_d t + \theta)\right)u(t)$$

Derivation:
Transfer function:1M
Partial fraction method:A=1, B=-1, C=-1 4M
Inverse Laplace Transform:3M
Final output C(t):2M

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SCRUTINY & EVALUATION OF CIE QUESTION PAPER

Declaration by the Course handling faculties:

As a teaching faculty of the course _____ with code: _____,
We hereby confirm that the question paper is thoroughly reviewed and we ensure that it adheres to the following criteria:

Sl. No	Criteria	Yes/NO
1	The question paper adequately covers the prescribed syllabus contents.	
2	The question paper is in line with the recommended pattern, taking into consideration the structure and format suitable for the Continuous Internal Evaluation.	
3	The question paper is designed to align with the Revised Bloom's Taxonomy, encompassing various levels of cognitive skills such as remembering, understanding, applying, analyzing, evaluating, and creating.	
4	The question paper is aligned with the defined course outcomes, ensuring that it effectively assesses the knowledge and skills acquired during the course.	
5	Course handling faculty (As applicable) are responsible for preparing the question paper, scheme, and solution have unanimously agreed to utilize this question paper for conducting the Continuous Internal Evaluation.	
6	The question paper, Scheme and Solution has been submitted to the Test coordinators within the designated timeframe to ensure the smooth conduction of Continuous Internal	

Course handling Faculties:

Name: **Signature**

1.
2.
3.
4.

To be filled by the Scrutinizer:

Sl. No	Rubrics	Points	
		Max	Awarded
1	Timely submission of the question paper along with the scheme & solution	10	
2	Heterogeneous nature of QP with respect to BTs and Cos	10	
3	Format with proper entry of all particulars including test, course name, code, date, max marks, BT CO table, efficient use of paper (proper spacing, figures)	10	
4	No handwritten data or diagrams	10	
5	Scheme & Solution in the format	10	
	Total points	**50**	
Any other comments by the scrutinizer :			

Note: Course coordinators to obtain scrutinizer's acceptance by incorporating all suggestions from scrutiny into the final versions of QP, Scheme, and Solutions.

All corrections suggested by the scrutinizer are incorporated and both the copies are re-submitted	**Signature of Course coordinator**
Accepted/Rejected	**Signature of Scrutinizer** **(Name:)**

Signature of HOD